Curriculum

Embedded Systems and Internet of Things



Module	Topics Covered
IoT Architecture & Ecosystem	IoT layers, protocols (MQTT, HTTP, CoAP), IoT vs IIoT, real-world applications
Embedded C & Microcontrollers	Embedded C basics, GPIO, timers, interrupts, memory mapping
Arduino Programming	Arduino IDE, digital/analog I/O, PWM, serial communication, sensor interfacing
Raspberry Pi & Linux Basics	Pi setup, Linux terminal, Python on Pi, SSH, GPIO with RPi.GPIO
Sensors & Actuators	Temperature (DHT11), motion (PIR), ultrasonic, LDR, relay, servo – hands-on
IoT Communication Protocols	MQTT with Mosquitto, WiFi (ESP8266/ESP32), Bluetooth, LoRa intro
Cloud Integration	ThingSpeak / AWS IoT / Firebase – sending sensor data to cloud, dashboard creation
IoT Capstone Project	Smart Home / Smart Agriculture / Industrial Monitor – full end-to-end prototype